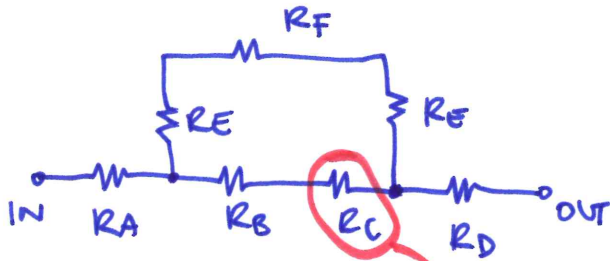
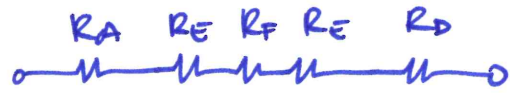


PART 1

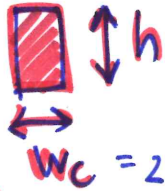


NO CELL



CELL

$$\frac{W}{h} = \frac{50}{10} = 5 > 1.4$$



$$\frac{h}{W_c} = 5 > 1.4$$

$$\beta = \frac{12 \cdot 10^{-3}}{(1 - 0.63 \frac{1}{5}) (2 \cdot 10^{-6})^3 \cdot 10 \cdot 10^6} = 1.71 \cdot 10^{20}$$

$$\alpha = \frac{12 \cdot 10^{-3}}{(1 - 0.63 \frac{h}{W}) h^3 \cdot W} = \frac{12 \cdot 10^{-3}}{(1 - 0.63 \frac{1}{5}) (10 \cdot 10^{-6})^3 \cdot 50 \cdot 10^{-6}} = 2.74 \cdot 10^{17}$$

$$R_A = \alpha \cdot 1 \text{ m} = 2.74 \cdot 10^{14} \text{ [Pa} \cdot \text{s/m}^3]$$

$$R_B = 2 \cdot R_A = 5.49 \cdot 10^{14}$$

$$R_D = R_A = 2.74$$

$$R_E = R_B = R_F = 5.49$$

$$R_C = \beta \cdot 20 \cdot 10^{-6} = 3.43 \cdot 10^{15}$$

$$\text{NO CELL} \quad R_A + 3R_E \parallel (R_B + R_C) + R_D = 1.7 \cdot 10^{15}$$

$$\text{CELL} \quad R_A + 3R_E + R_D = 2.19 \cdot 10^{15} \rightarrow \text{WORST CASE}$$

$$R_{\max} = \frac{5 \text{ atm}}{4 \mu\text{e/min}} = \frac{5 \cdot 101 \text{ kPa}}{4 \cdot 10^{-9} \text{ m}^3 / 60 \cdot 5} = 7.57 \cdot 10^{15}$$

$$\# = \frac{R_{\max}}{R_{\text{cell}}} = \frac{7.57}{2.19} = [3.45] = \underline{3 \text{ DEVICES}}$$

$$\text{from figure } v \sim 5 \frac{\text{mm}}{\text{s}}$$

$$Q = v \cdot h \cdot W =$$

$$= 5 \cdot 10^{-3} \frac{\text{m}}{\text{s}} \cdot 10 \mu\text{m} \cdot 50 \mu\text{m} =$$

$$= 2.5 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}} = \underline{0.15 \mu\text{e/min}}$$

↓) IF LAMINAR FLOW

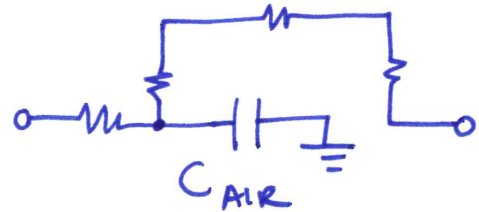
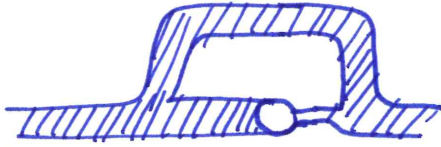
$$R_{IN} \rightarrow \text{---} R_{OUT}$$

$$R_{IN} = R_{OUT}$$

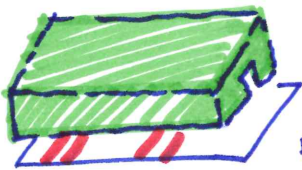
∴) BLOOD $\Rightarrow \eta_{BLOOD} = 4 \div 5 \cdot \eta_{WATER} \Rightarrow R$ INCREASES

• AIR

↓
COMPRESSIBLE $\Rightarrow$ CAPACITANCE



1) SUBSTRATE
LIFT-OFF of ELECTRODES ON GLASS
1) SPIN COAT...

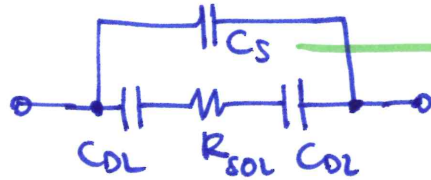
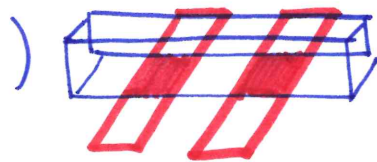


PDM CHANNEL

PLASMA BONDING

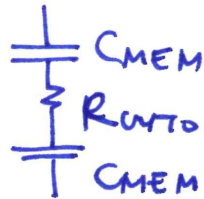
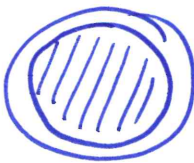
- 1) MASTER
- 2) POUR (MIX)
- 3) DEGAS
- 4) CURE
- 5) PEEL-OFF...

PART 2



NOT VISIBLE
IN SPECTRUM

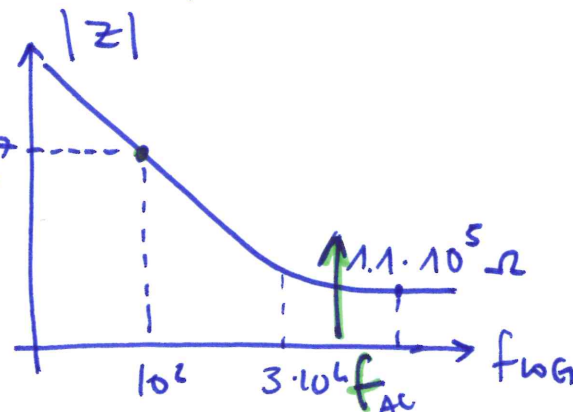
2) CELL:



INSULATING
SPHERE

INCREASE
 R_{SOL}

3) from plot



$$f_{OPT} > 3 \cdot 10^4$$

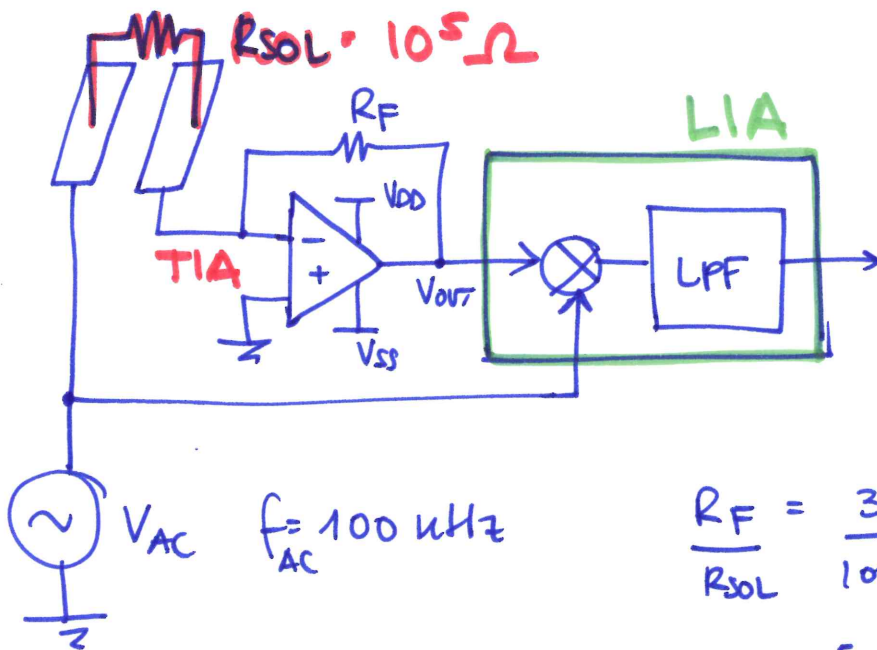
$\sim 100 \text{ kHz}$

AREA

$$C = \frac{1}{2\pi \cdot 10^2 \cdot 3 \cdot 10^7} \approx 53 \text{ pF} = \frac{C_{DL}}{2} \rightarrow C_{DL} = 106 \text{ pF} = \frac{0.1 \text{ pF}}{\mu\text{m}^2} \cdot W \cdot L$$

$$WL = 1060 \mu\text{m}^2 \quad L = \frac{1060 \mu\text{m}^2}{50 \mu\text{m}} = 21 \mu\text{m}$$

j)



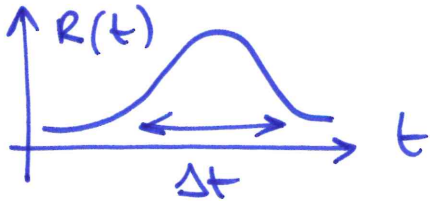
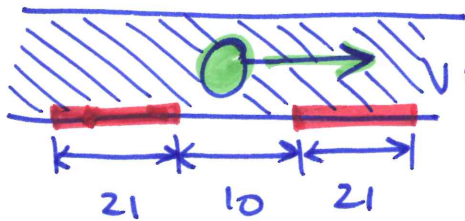
$$1 \mu A \quad V_{AC} = 100 \text{ mV}$$

$$\frac{R_F}{R_{SOL}} = \frac{3V}{100 \text{ mV}}$$

$$R_F = 10^5 \Omega \cdot \frac{3}{0.1} = 3 \text{ M}\Omega$$

Check $BW = \frac{1}{2\pi R_F \cdot 0.2 \text{ pF}} = 2.65 \text{ kHz}$ (a bit close, $1 \text{ M}\Omega$ would be better)
 $> f_{AC}$

i)

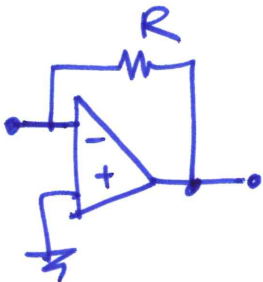


$$\Delta t = \frac{g}{v} = \frac{10 \mu\text{m}}{33 \text{ mm/s}} \approx 300 \mu\text{s}$$

(WORST CASE SHORTEST)

$$BW_{LIA} \approx \frac{1}{\Delta t} \sim \frac{1}{300 \mu\text{s}} = 3.3 \text{ kHz}$$

)



$$S_i = \frac{4kT}{R}$$

$$\sigma_{IN} = \sqrt{\frac{4kT}{R} \cdot \frac{1}{2\pi R \cdot C} \cdot \frac{\pi}{2}} = 100 \text{ pA}$$

S_i
 BW
 $C \sim 0.2 \text{ pF}$

$$100 \text{ pA} = \frac{1}{R} \cdot \sqrt{\frac{kT}{C}}$$

$$R = \frac{1}{100 \text{ pA}} \cdot \sqrt{\frac{kT}{C} \cdot \frac{9}{26 \text{ mV}}} = 1.4 \text{ M}\Omega$$

TIA ONLY
(NO LIA FILTER)